

Secure Train Ticketing with Transaction Management

Project Details

--> Tech Stack: Java, SQL

--> Timeline: 1 Month

--> Team Size: 1 Member

REQUIREMENT GATHERING & ANALYSIS

1. Login Feature
2. User/Admin Account Creation
3. Role-Based Access
4. Train Scheduling
5. Seat Assignment
6. Live Booking Status
7. Passenger Directory
8. Transaction Time Tracking
9. Booking Listing
10. Ticket Priority
11. Notification.
12. Asset Management: Managing train coaches and total seat capacity.
13. Report Management System: Generating daily collection reports.
14. Journey Planning: Scheduling and coordinating train departures.
15. File Management System: System to manage electronic ticket records.
16. Support Feature: Feedback or support communication feature for users.
17. Budget/Refund Tracking: Monitoring transaction reversals and revenue.

FEASIBILITY ANALYSIS

10. Ticket Priority/Class Set: Yes - Included in current phase.
15. File Management System: No - Next phase.
16. Support/Chatting Feature: No - Next phase.
8. Transaction Time Tracking: Yes.
12. Train Asset Management: Yes.
14. Journey Planning: No - Standard scheduling only.

FEATURES

1. SECURITY & ACCESS

- User account creation.
- Role-based access.
- Login feature.

2. ADMIN

- All passenger details view option.
- Train & Seat assignment.
- Status of occupancy.
- Report of daily transactions.

3. PASSENGER

- My Ticket list.
- Current / Completed / Cancelled tickets.
- Status update of booking.
- Notification for passengers.

PROJECT: SECURE TRAIN TICKETING WITH TRANSACTION MANAGEMENT

1. PROJECT DETAILS

- * Tech Stack: Java, SQL
- * Timeline: 1 Month
- * Team Size: 1 Member

2. FEASIBILITY ANALYSIS & SCOPE

Included in Current Phase:

- * Ticket priority classes (AC, Sleeper, General).

- * Transaction time tracking.
- * Train asset management (coaches, capacity).

Moved to Next Phase:

- * File management system.
- * Support or chatting feature.
- * Advanced journey planning (restricted to standard scheduling only).

3. CORE FEATURES

SECURITY & ACCESS

- * Secure login feature.
- * User account creation for new passengers and staff.
- * Role-based access with different permissions for Admin and Customer.

ADMIN PORTAL

- * View option for all registered passenger details.
- * Train scheduling and seat assignment capabilities.
- * Real-time status updates of seat availability and occupancy.
- * Report generation for daily transactions and revenue.

PASSENGER PORTAL

- * View of all tickets booked under a specific user.
- * Track current, completed, and cancelled tickets.
- * Status update of active bookings.
- * Notifications for successful bookings or cancellations.

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SYSTEM MODELS (Database & Application Entities)

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1. USER MODEL

id: Long
userId: String
name: String
email: String
password: String (hashed)
mobileNo: String
dob: Long

role: Enum - Admin, Customer
status: Enum - Active, Inactive
createdAt: Long

2. TRAIN MODEL

id: Long
trainNumber: String
trainName: String
sourceStation: String
destinationStation: String
totalCoaches: Integer
totalSeatCapacity: Integer
status: Enum - Active, Inactive

3. SCHEDULE MODEL

id: Long
trainId: Long (FK - Train.id)
journeyDate: Long
departureTime: Long
arrivalTime: Long
status: Enum - Scheduled, Departed, Arrived, Cancelled

4. TICKET MODEL

id: Long
pnrNumber: String
scheduleId: Long (FK - Schedule.id)
bookedBy: Long (FK - User.id)
passengerId: Long (FK - User.id)
ticketClass: Enum - AC, Sleeper, General
seatNumber: String
bookingTime: Long
updatedAt: Long
status: Enum - Current, Completed, Cancelled

5. TRANSACTION MODEL

id: Long

ticketId: Long (FK - Ticket.id)
userId: Long (FK - User.id)
amount: Double
transactionType: Enum - Booking, Refund
transactionDate: Long
status: Enum - Success, Failed, Refunded

6. NOTIFICATION MODEL

id: Long
userId: Long (FK - User.id)
ticketId: Long (FK - Ticket.id)
message: String
type: Enum - BookingSuccessful, Cancelled, RefundInitiated
isRead: Boolean
createdTime: Long

7. LOGIN REQUEST MODEL

email: String
password: String

GitHub link

<https://github.com/ssathish441/thodar-train-booking.git>